
DSC 140A - Homework 07

Homework 07 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using $\text{\LaTeX}$ with our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, convert them into PDF, and then merge the PDF with your math problems solutions generated in $\text{\LaTeX}$.

Problem 1.

The file linked below contains a data set of 150 samples. The first column contains a single continuous feature, X , assumed to have been drawn from an unknown probability density. The second column contains the binary class label Y .

https://f000.backblazeb2.com/file/jeldridge-data/011-univariate_density_estimation/data.csv

In the first problem on this homework, you used a histogram estimator to apply the Bayes classification rule. In this problem, you will instead estimate the class-conditional densities by fitting Gaussians using the method of maximum likelihood. In each part, show your code and provide your reasoning.

Note: you should *not* use `sklearn`, `scipy`, or any other library to fit the Gaussian densities. You should instead calculate the maximum likelihood estimates directly (using `numpy` or `pandas` to do this is fine).

- a) Estimate the class-conditional densities $p_X(x|Y=0)$ and $p_X(x|Y=1)$ with Gaussians using the method of maximum likelihood and report the estimated parameters.

Solution: To estimate the class-conditional densities, we separate the data with label 0 and the data with label 1 and estimate the parameters μ and σ separately for each subset using the MLE formulas. The MLE for the mean is the sample mean, and the MLE for the variance is the sample variance. The code that does this is:

```
data = np.loadtxt('data.csv', delimiter=',')
y = data[:, 1]
X_1 = data[y == 1, 0]
X_0 = data[y == 0, 0]

# "fit" a Gaussian distribution to the data
mu_1 = np.mean(X_1)
sigma_1 = np.std(X_1)

mu_0 = np.mean(X_0)
sigma_0 = np.std(X_0)
```

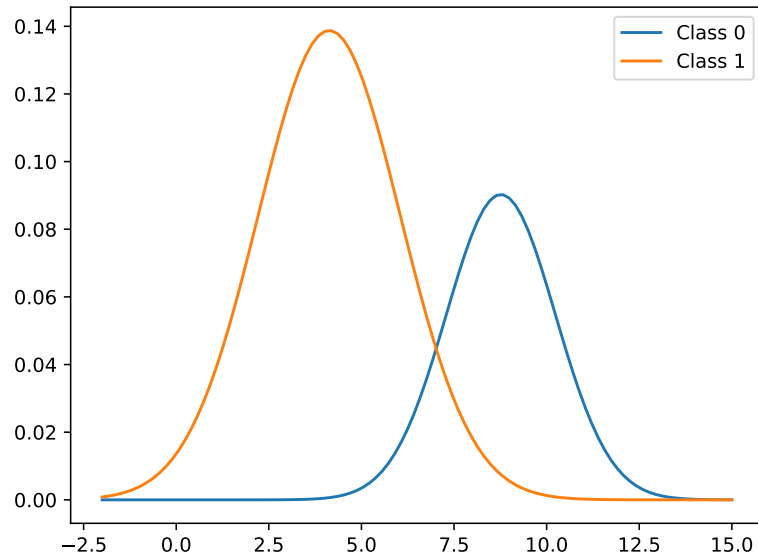
The parameter estimates we get are:

Parameter	Value
μ_1	4.13
σ_1	1.91
μ_0	8.76
σ_0	1.47

- b) Let $\tilde{p}_X(x|Y=0)$ and $\tilde{p}_X(x|Y=1)$ be the estimated class-conditional densities from the previous part, and let $\tilde{\mathbb{P}}(Y=1)$ be the estimate for $\mathbb{P}(Y=1)$.

Plot $\tilde{p}_X(x|Y=0) \cdot \tilde{\mathbb{P}}(Y=0)$ and $\tilde{p}_X(x|Y=1) \cdot \tilde{\mathbb{P}}(Y=1)$ on the same axis. Label your plot so that the grader can tell which Gaussian corresponds to which class. Remember to show your code.

Solution:



```
def make_gaussian(mu, sigma):
    """Returns a function that evaluates a Gaussian density."""
    def f(x):
        return 1/np.sqrt(2 * np.pi * sigma**2) * np.exp(-(x - mu)**2 / (2 * sigma**2))
    return f

g1 = make_gaussian(mu_1, sigma_1)
g0 = make_gaussian(mu_0, sigma_0)

# make functions which compute p(x/Y) P(Y)

def s1(x):
    return g1(x) * np.mean(y)

def s0(x):
    return g0(x) * (1 - np.mean(y))

xx = np.linspace(-2, 15, 100)

plt.plot(xx, g0(xx), label='Class 0')
plt.plot(xx, g1(xx), label='Class 1')
plt.legend()

plt.savefig('../include-solution/part_b.pdf')
```

- c) Suppose a new point is observed at $x = 6.271$. What class does the Bayes classifier predict for x when the estimated densities and probabilities are used? Remember to show your code, and determine the predicted class through calculation (and not by visual inspection of the plot from the previous part).

Solution: To determine the prediction of the Bayes classifier, we calculate:

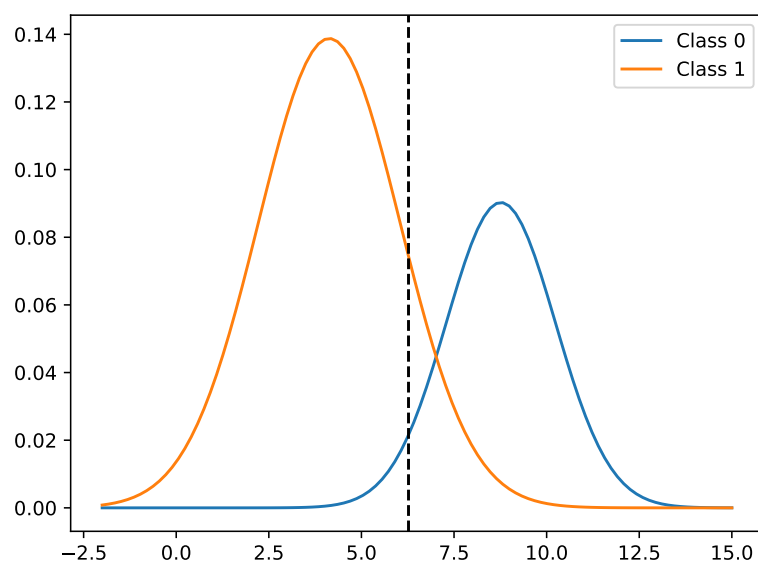
$$p(x | Y = 0) \cdot \mathbb{P}(Y = 0) \quad \text{and} \quad p(x | Y = 1) \cdot \mathbb{P}(Y = 1)$$

In our code above, we already created functions `s0` and `s1` that calculate these values. We can use these functions to calculate the predictions for $x = 6.271$:

```
print(s0(6.271))
print(s1(6.271))
```

For Class 0, we get 0.022, while for Class 1 we get 0.074. Thus, the Bayes classifier predicts that $x = 6.271$ belongs to Class 1.

Note that we could have guessed this from the plot from part (b). If we draw a dashed line at $x = 6.271$, we can see that $p(6.271 | Y = 1) \cdot \mathbb{P}(Y = 1)$ is larger than $p(6.271 | Y = 0) \cdot \mathbb{P}(Y = 0)$:



- d) The `scipy.optimize.fsolve` function can be used to find the *roots* of a function f ; that is, the places where function $f(x) = 0$. Use `fsolve` to find the decision boundary for the classifier you trained above. Show your code.

Note that there may actually be multiple decision boundaries at the extremes, but there is one clear decision boundary in the middle of the data, as should be evident in your plot; find that one.

Solution: We use `scipy.optimize.fsolve` to find the decision boundary. It searches for the *roots* of a function, f . To use it to find where the decision boundary is, we define the function f :

$$f(x) = p(x | Y = 1) \cdot \mathbb{P}(Y = 1) - p(x | Y = 0) \cdot \mathbb{P}(Y = 0)$$

This will be zero precisely when $p(x | Y = 1) \cdot \mathbb{P}(Y = 1) = p(x | Y = 0) \cdot \mathbb{P}(Y = 0)$, which is exactly the decision boundary.

In code:

```
import scipy.optimize

def f(x):
```

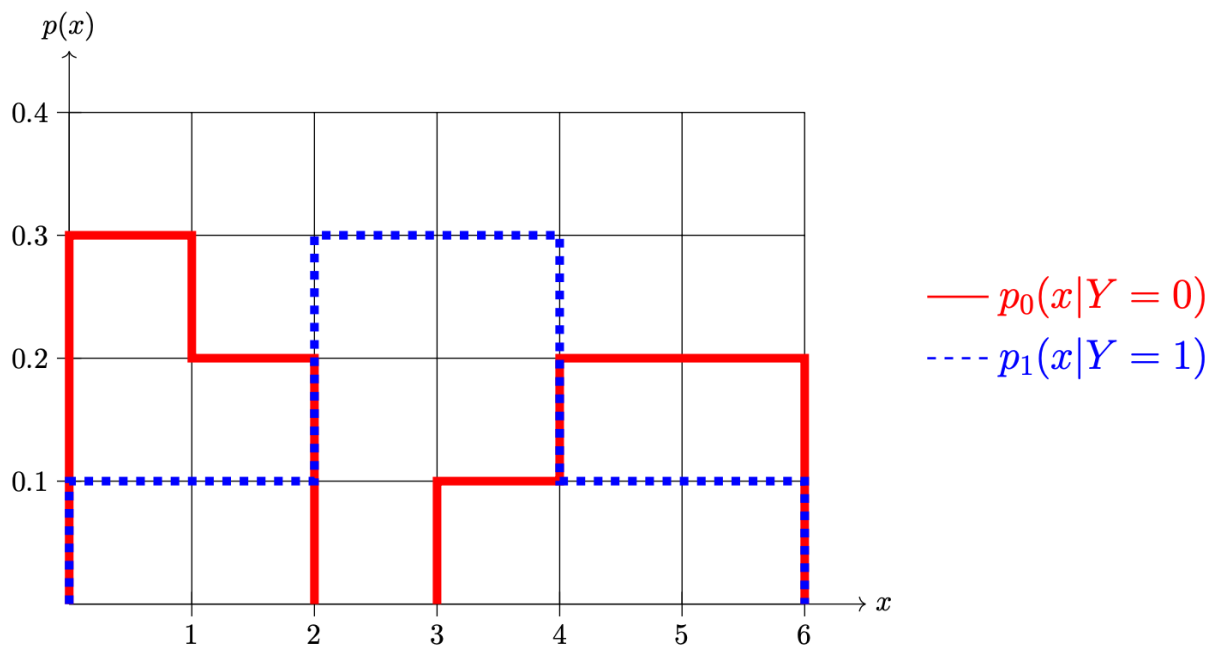
```
return s1(x) - s0(x)
```

```
scipy.optimize.fsolve(f, x0=5)
```

This gives 7.01. Referencing our plots above, this looks to be where the orange and blue curves intersect, and is therefore reasonable.

Problem 2.

Shown below are two conditional densities, $p_1(x|Y=1)$ and $p_0(x|Y=0)$, describing the distribution of a continuous random variable X for two classes: $Y=0$ (the solid line) and $Y=1$ (the dashed line). You may assume that both densities are piecewise constant.



- a) Suppose that $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is $\mathbb{P}(1 \leq X \leq 3)$?

Solution: 0.3

- b) Suppose $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is the prediction of the Bayes classifier at $x = 1.5$?

Solution: Class 0.

- c) Suppose again that $\mathbb{P}(Y=1) = \mathbb{P}(Y=0) = 0.5$. What is the Bayes error with respect to this distribution?

Solution: 0.25

- d) Now suppose $\mathbb{P}(Y=1) = 0.7$ and $\mathbb{P}(Y=0) = 0.3$. What is $\mathbb{P}(1 \leq X \leq 3)$?

Solution: 0.34

- e) Suppose again that $\mathbb{P}(Y=1) = 0.7$ and $\mathbb{P}(Y=0) = 0.3$. What is the prediction of the Bayes classifier at $x = 1.5$?

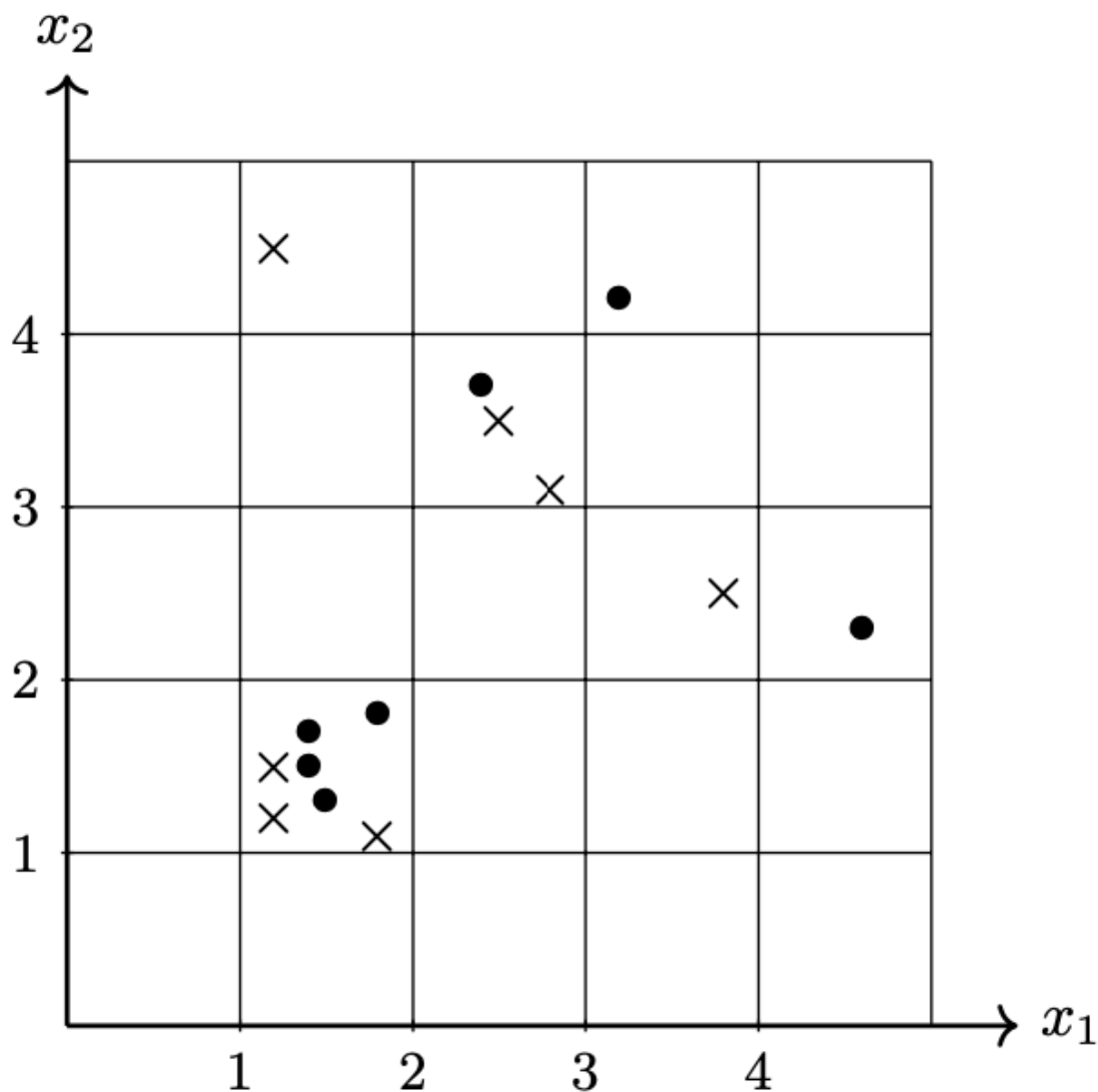
Solution: Class 1.

- f) Suppose again that $\mathbb{P}(Y = 1) = 0.7$ and $\mathbb{P}(Y = 0) = 0.3$. What is the Bayes error with respect to this distribution?

Solution: 0.28

Problem 3.

Consider the following data set of 14 points in $\mathbb{R}^2$. Each point has a label in $\{1, -1\}$. Points from Class 1 are marked with $\times$ and points from Class -1 are marked with $\bullet$.



Suppose the class-conditional density estimates are computed from this data using a histogram density estimator using the 1×1 bins shown in the figure above.

If these estimates are used in place of the true class-conditional densities in the Bayes classifier, what will be the training error of the classifier? That is, what percentage of the data above will be misclassified? You

may leave your answer as a fraction or a decimal.

Solution: $4/14 = .2857$